

The Gyromagnetic Movement Mechanism: Rotation as Accumulated Torque, Helicity as Invariant, and the Elsasser Attractor

TZPID Gold Spine Series, Paper IV of VIII

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June 2026

Abstract

Papers I–III of this series established the geometry of nested hyperspherical enclosures, the phase-locking mechanism that selects rational ratios on them, and the holonomy signature of their curvature. None of it yet moves anything. This paper supplies the engine. Drawing on three source treatments—the Vimana gyromagnetic analysis (local state-space and field skeleton), *Our Gyromagnetic Universe* (the cross-scale movement law), and the gyromagnetic validation study (categorical and metric bridges)—we assemble a single movement mechanism in eight links: configuration space $T^*(SO(3) \times \mathbb{R}^3) \otimes \mathfrak{u}(1)$; compressed-dipole effective moment; rotating metric correction $g_{\mu\nu}[\omega] = \eta_{\mu\nu} + h_{\mu\nu}^{\text{rot}}$; vector-potential/helicity vortex with the topological invariant $H_M = \int_V \mathbf{A} \cdot \mathbf{B} dV$; the Elsasser attractor $\Lambda = B^2/(\rho\mu\Omega^2) \rightarrow 1$ with magnetic–Coriolis equipartition; the lab-to-astro scaling functor preserving dimensionless invariants ($\mu = kMR^2\Omega$); the magnetoacoustic gateway $\delta\Lambda(\omega) = \alpha U(\omega)/(\kappa + i\omega)$ by which cavity modes drive the system; and orbital phase locking as the terminal capture. The central conceptual inversion is the registry’s Emergent Rotation Principle (ID0037, ID0085, ID0087): angular momentum is not a primordial endowment but a continuously generated quantity, $\omega(t) = \omega(0) + I_T^{-1} \int_0^t \tau_{\text{em}} dt'$ —movement is *accumulated*, not inherited. Wolfram-certified guardrails verify torque orthogonality, Elsasser relaxation $\dot{\Lambda} = -\kappa(\Lambda - 1)$, equipartition, helicity constancy, the gyromagnetic scaling ratio, and the vanishing of rotating-metric and gravitomagnetic corrections in the $\omega \rightarrow 0$ limit. The accumulation principle established here is generalized to gravitation itself in Paper V.

1 Introduction: from kinematics to a movement engine

The first three papers of this series were deliberately bloodless. They concerned spectra (Paper I), locks (Paper II), and holonomies (Paper III)—the kinematic skeleton of the nested-enclosure framework. The present paper asks the dynamical question: what actually *moves* matter through the enclosure, and why is so much of the universe’s visible structure—planets, stars, accretion disks, galaxies—*rotating*?

The conventional answer treats rotation as a remnant: primordial angular momentum, conserved and redistributed. The TZPID registry asserts a different and falsifiable position, the *Emergent Rotation Principle*: rotation is continuously generated by electromagnetic torque density and accumulated over history. Three registry entries state it with increasing operational force. ID0037 (Core entry, “Emergent Rotation Principle”):

$$L = \int_V (\mathbf{r} \times \mathbf{J}) d^3x, \quad \frac{dL}{dt} = \int_V (\mathbf{J} \times \mathbf{B}) d^3x, \quad (1)$$

angular momentum driven by local fields, not inherited. ID0085 (“Gyromagnetic Genesis Hypothesis”) adds the full torque density,

$$\boldsymbol{\tau}_{\text{em}} = \int \mathbf{r} \times (\mathbf{J} \times \mathbf{B} + \rho_e \mathbf{E}) dV, \quad \frac{d\mathbf{L}}{dt} = \boldsymbol{\tau}_{\text{em}}, \quad (2)$$

and ID0087 (“Rotation as a Continuously Generated Quantity”) makes the accumulation explicit:

$$\dot{\omega}(t) = \frac{\tau_{\text{em}}(t)}{I_T}, \quad \omega(t) = \omega(0) + \frac{1}{I_T} \int_0^t \tau_{\text{em}}(t') dt'. \quad (3)$$

Equation (3) is the sentence the rest of the series conjugates: *motion is the integral of its own history*. Paper V will write the same sentence for gravity (acceleration corrected by accumulated mass transport); Paper VI for curvature (a causal accumulation kernel over effective stress); Paper VII for matter itself (vacuum pressure accumulating to a creation threshold). The gyromagnetic mechanism is where the accumulation principle is closest to laboratory test, which is why it sits at the hinge of the series.

The mechanism is assembled as a chain:

configuration space → compressed-dipole moment → rotating metric correction
→ helicity vortex → Elsasser attractor → lab-to-astro functor
→ acoustic boundary drive → orbital phase locking,

each link anchored to extracted TeX witnesses from the three source treatments and bridged to registry obligations. Within the series’ formal pipeline this paper is the declared pre-Isabelle dependency layer of Paper II: the moving gyromagnetic system and its invariants must be typed before the phase-lock and rational-selection obligations are proved.

2 Link 1: the state space

The Vimana analysis supplies the configuration-space skeleton (witness `vimana_config_space`; registry ID10146, `Derived_Theorem_Obligation`):

$$\mathcal{M} = T^*(SO(3) \times \mathbb{R}^3) \otimes \mathfrak{u}(1)_{\text{gauge}}, \quad \Psi(t) = (r_{\text{cm}}, \omega_{\text{gyro}}, \mu_{\text{constrained}}, L_{\text{mech}})^T, \quad (4)$$

the cotangent bundle of rigid placement and orientation, tensored with a gauge phase. Two registry entries fix its symmetry and conservation structure. ID0092 (`Core_Definition`) gives the interacting symmetry group as a semidirect product,

$$G = SO(3) \ltimes_{\varphi} U(1), \quad (R, e^{i\alpha}) \cdot (R', e^{i\beta}) = (RR', e^{i\alpha} \varphi_R(e^{i\beta})), \quad (5)$$

rotation acting on gauge phase—the algebraic statement that orientation and electromagnetic phase are coupled sectors, not adjacent labels. ID0093 (`Core_Definition`) imposes symplectic preservation,

$$\phi_t^* \omega = \omega, \quad \mathcal{L}_{X_H} \omega = 0 : \quad (6)$$

admissible gyromagnetic flows are symplectomorphisms and conserve the Liouville measure. The state space is thus typed before any dynamics. This is the first pre-Isabelle formalization target, `gyro_state_space`; its Wolfram guardrail, `config_state_dimension`, certifies the dimension count of (4).

3 Link 2: the compressed-dipole effective moment

The Vimana source’s opposing-dipole configuration (witness `vimana_opposing_dipoles`; registry bridges ID0044, ID0045, ID10131) carries the effective moment

$$\mu_{\text{eff}} = \mu_1 + \mu_2 + \frac{\hbar}{2m_e c} \frac{\nabla \times (\mu_1 \times \mu_2)}{4\pi\epsilon_0 c^2 d(t)^3}, \quad (7)$$

a base dipole sum plus a short-distance correction that grows as the separation $d(t)$ is compressed. In the formal pipeline the physical claim attached to (7) is deliberately guarded as a hypothesis (`effective_dipole`); what is consumed downstream is only its structure: a constrained moment $\mu_{\text{constrained}}$ entering the state vector (4). The celestial diagnostic associated with it is the gyromagnetic ratio (ID10131)

$$\gamma = L/\mu, \quad (8)$$

certified as `gyromagnetic_ratio`, which Section 7 promotes to a cross-scale observable. A topological constraint completes the link: ID10130 (`Derived_Theorem_Obligation`) invokes the S^3 universe topology and the Hopf map $\eta : S^3 \rightarrow S^2$ to block flat dipole collapse—the same projection family that carried the comma in Paper III, now acting as a structural guard on field configurations.

4 Link 3: the rotating metric correction

Rotation back-reacts on geometry. The Vimana extraction (witness `vimana_rotating_metric`; bridges ID9940, ID10145, ID10272) writes the rotating correction

$$g_{\mu\nu}[\omega] = \eta_{\mu\nu} + h_{\mu\nu}^{\text{rotation}}, \quad h \sim c^{-2} [(\omega \times r)^2, -(\omega \times r)], \quad (9)$$

the frame-dragging analogue controlled by powers of the rim speed $\omega R/c$. Two registry obligations calibrate the regimes: ID10145, the laboratory regime $\omega R \sim 10^{-4}c$ where the correction is real but minute, and ID10272, the relativistic regime $\omega R \rightarrow c$ where gravitomagnetic coupling turns on, with the source-integrated field

$$B_{\text{gravitomagnetic}} = \frac{2G}{c^2} \frac{\omega \times r}{r^3} \int_V \rho(r') |r'|^2 d^3 r' \quad (10)$$

(witness `vimana_gravitomagnetic_coupling`). The formal guardrails `rotating_metric_zero_limit` and `gravitomagnetic_zero_limit` certify that both corrections vanish identically at $\omega = 0$ and are controlled by $\omega R/c$ —the pre-Isabelle target `rotating_metric_limit`. The observable bridge entry ID10264 assembles the testable suite: frame-dragging phase $\Delta\phi = (4GJ_{\oplus}/c^2 r) \int_0^T \sin^2 \theta dt$, Kaluza–Klein resonance shift $\delta\omega/\omega = m_{\text{KK}}c/(R_{\text{KK}}\omega_0)$, accumulated-field ratio $g_{\text{acc}}/g_{\text{Newton}} = 1 + \alpha t + \beta t^2$, and helicity transfer $\mathcal{H}_{\text{mag}} \rightarrow \mathcal{H}_{\text{KK}}$. The third item is the direct forerunner of Paper V’s accumulated-force law.

5 Link 4: the helicity vortex and its invariant

The movement mechanism’s conserved quantity is topological. The Vimana vector-potential extraction (witness `vimana_vector_potential_helicity`; bridges ID9758, ID9760, ID9761, ID10760) decomposes the total potential and defines the helicity

$$A_{\text{total}} = A_{\text{dipole}} + A_{\text{gyro}} + A_{\text{induced}}, \quad H(t) = \int \mathbf{A} \cdot \mathbf{B} d^3 r - \frac{\omega_{\text{gyro}}}{c^2} \int \mathbf{r} \times (\mathbf{E} \times \mathbf{B}) d^3 r, \quad (11)$$

whose leading term is the registry’s movement invariant (ID9758, Invariant):

$$H_M = \int_V \mathbf{A} \cdot \mathbf{B} dV. \quad (12)$$

Magnetic helicity measures the linking and twisting of field lines. It is the volumetric realization of the Gauss linking number that Paper VIII will use as a building block of the unified topological charge. Under ideal deformation it is conserved; the guardrail is `helicity_constant`, and the pre-Isabelle target is `helicity_invariant`. The rotating-frame equilibrium condition (ID9761) closes the force balance:

$$\int_V (\mathbf{J} \times \mathbf{B} - \rho \boldsymbol{\Omega} \times \mathbf{v}) dV = 0, \quad (13)$$

Lorentz force against Coriolis load. Helicity plays in this paper the role the comma played in Paper III: the loop-level invariant that survives all continuous rearrangement, and therefore the bookkeeping quantity for movement through the enclosure.

6 Link 5: the Elsasser attractor

Our Gyromagnetic Universe supplies the cross-scale balance point (witness `ogu_elsasser_manifold`; registry ID0038, ID0039, ID0040). Define the Elsasser-type parameter and its relaxation law:

$$\Lambda = \frac{B^2}{\rho \mu \Omega^2}, \quad \dot{\Lambda} = -\kappa(\Lambda - 1) + \chi \delta S(t). \quad (14)$$

ID0038 declares the attractor manifold $\Lambda \approx 1$; the guardrail `elsasser_relaxation` certifies that the normal form relaxes to $\Lambda = 1$ for $\kappa > 0$ (pre-Isabelle target `elsasser_relaxation`). At the attractor, magnetic and rotational energy densities equipartition (ID0039, certified as `equipartition_identity`):

$$\frac{B^2}{2\mu} = \frac{1}{2} \rho (\Omega r)^2, \quad E_{\text{mag}} = E_{\text{rot}}, \quad (15)$$

and ID0040 asserts universality: $\Lambda_{\text{crit}} \approx 1$ across laboratory and astrophysical systems regardless of scale, deviations bounded by $O(1)$ factors. The Elsasser attractor is the dynamical analogue of the critical surfaces of Paper I, Pillar 3: a self-organizing balance to which rotating magnetized systems relax, indexed not by material but by dimensionless ratio—the reason a laboratory experiment can speak to a planetary dynamo at all.

7 Link 6: the lab-to-astro scaling functor

That reason is made rigorous categorically. The scaling extraction (witness `ogu_scaling_functor`; bridges ID0044, ID0089, ID0090) records the empirical scaling laws

$$B \propto \sqrt{\rho} L^{1/2} \Omega, \quad \mu \propto M R^2 \Omega, \quad (16)$$

the second being the registry’s universal magnetic-moment law (ID0044). Its corresponding certificate records the gyromagnetic scaling ratio:

$$\mu = k M R^2 \Omega, \quad k \approx \text{const.} \quad (17)$$

ID0089 (Core_Definition) elevates the comparison to a functor between categories,

$$F : \mathbf{LabGyro} \longrightarrow \mathbf{AstroRot}, \quad F(g \circ f) = F(g) \circ F(f), \quad (18)$$

preserving objects (configurations) and morphisms (dynamical evolutions), with natural transformations $\eta : F \Rightarrow G$ expressing equivalence up to scale. ID0090 states the admissibility condition: the scale map must preserve the dimensionless invariants,

$$\Pi_k^{\text{lab}} = \Pi_k^{\text{astro}}, \quad F_{\text{scale}} : \mathcal{S}_{\text{lab}} \rightarrow \mathcal{S}_{\text{astro}}, \quad (19)$$

Reynolds and magnetic-Reynolds numbers among them. The methodological content is sharp: *laboratory demonstrations count as structural evidence for astrophysical dynamics only when connected by a verified functor preserving* (19), *never by loose analogy* (pre-Isabelle target `lab_astro_functor`). This is the same discipline Paper I imposed on the ripple data—dimensionless form conserved, absolute scale set by the medium—now stated as a category-theoretic admissibility rule.

8 Links 7–8: the acoustic gateway and orbital capture

The enclosure re-enters as the drive. The OGU magnetoacoustic-gateway witness, bridged through ID0252, ID9513, and ID10018, gives the response of the Elsasser parameter to acoustic forcing at frequency ω :

$$\delta\Lambda(\omega) = \frac{\alpha U(\omega)}{\kappa + i\omega}, \quad (20)$$

a first-order low-pass response. The cavity modes of the Well/Wall/Wave selector (ID0252, Paper II) pump the gyromagnetic state, and the relaxation rate κ of (14) sets the bandwidth. The guardrail is the magnetoacoustic response shape. The spin–field coevolution Hamiltonian structure, witnessed by the OGU spin–field Hamiltonian certificate with ID0037, ID0085, and ID0087, lives on $T^*SO(3) \times \mathbb{R}^n$ with torque $\tau = \mu \times B$. It closes the loop back to the accumulation law (3). Its torque-orthogonality guardrail certifies that $\mathbf{J} \times \mathbf{B}$ torques never do work along the field. Driven by (20) and dissipated tidally, the system’s terminal states are precisely the phase locks of Paper II: coupling defeats detuning at the threshold $K = |\omega_1 - \omega_2|$ (ID9513), and rational spin–orbit ratios are captured (ID0143, Mercury’s 3:2). The movement mechanism thus does not merely coexist with the ratio-selection mechanism; it is its physical substrate, encoded as the orbital-movement bridge.

The categorical validation layer, witnessed by `validation_gyro_topos` with bridges ID0089, ID0092, and ID10264, packages the whole mechanism as a pair of structure-preserving maps:

$$\begin{aligned} \text{FrameDragging} & : \mathbf{GyroscopicModes} \rightarrow \mathbf{SpacetimeMetric}, \\ \text{HelicityPreservation} & : \mathbf{MagneticStates} \rightarrow \mathbf{TopologicalInvariants}. \end{aligned} \quad (21)$$

with the effective field equation $G_{\mu\nu} = T^{\text{EM}} + T^{\text{top}}$ —electromagnetic plus topological stress as the source of geometry. That equation is the door through which Paper V walks.

9 The registry spine and proof order

Table 1 collects the registry bridge in proof order; the pre-Isabelle chain consumed by Paper II is

$$\begin{aligned} & \text{ID10146} \rightarrow \text{ID0037} \rightarrow \text{ID0087} \rightarrow \text{ID0038} \rightarrow \text{ID0039} \rightarrow \text{ID0044} \\ & \rightarrow \text{ID0089} \rightarrow \text{ID0090} \rightarrow \text{ID0252} \rightarrow \text{ID9513} \rightarrow \text{ID0143}. \end{aligned}$$

The seven pre-Isabelle formalization targets are ordered as follows:

1. `gyro_state_space`: type the components of Ψ .

Table 1: The gyromagnetic movement mechanism: registry bridge (abridged).

ID	Role in movement mechanism	Obligation	Verification
ID10146	Configuration space $T^*(SO(3) \times \mathbb{R}^3)$	Derived_Theorem	pass: state_dimension
ID0092	$SO(3) \ltimes_{\varphi} U(1)$ symmetry	Core_Definition	algebraic structure
ID0093	Symplectic preservation	Core_Definition	structural
ID0037	Emergent rotation $dL/dt = \int \mathbf{J} \times \mathbf{B}$	Semantic	pass: torque_orthogonality
ID0085	Gyromagnetic genesis hypothesis	Hypothesis	covered: torque_orthogonality
ID0087	Rotation as accumulated torque history	Hypothesis	pass: angular_velocity_integral
ID0038	Elsasser attractor $\Lambda \approx 1$	Semantic	pass: elsasser_relaxation
ID0039	Magnetic–Coriolis equipartition	Definition	pass: equipartition_identity
ID0040	Universality of Elsasser criticality	Semantic	covered: elsasser_relaxation
ID0044	Scaling law $\mu = kMR^2\Omega$	Semantic	pass: gyromagnetic_scaling
ID0089	FractalNodes functor $\text{Lab} \rightarrow \text{Astro}$	Core_Definition	category obligation
ID0090	Dimensionless-invariant preservation	Core_Definition	category obligation
ID9758	Helicity invariant $H_M = \int \mathbf{A} \cdot \mathbf{B} dV$	Invariant	pass: helicity_constant
ID9761	Rotating-frame force balance	Definition	field balance obligation
ID10130	S^3 topology blocks flat collapse	Derived_Theorem	topology obligation
ID10131	Gyromagnetic ratio $\gamma = L/\mu$	Derived_Theorem	pass: gyromagnetic_ratio
ID10145	Frame-dragging regime $\omega R \sim 10^{-4}c$	Derived_Theorem	pass: zero_limit
ID10272	Gravitomagnetic onset $\omega R \rightarrow c$	Derived_Theorem	pass: zero_limit
ID10264	Observable bridge suite	Derived_Theorem	observable suite
ID0252	Cavity modes as acoustic drive	Definition	covered: cavity_harmonic_2_1
ID9513	Phase-lock threshold	Definition	covered: phase_lock_threshold
ID0143	Spin–orbit 3:2 capture	Proposition	covered: spin_orbit_3_2

2. `effective_dipole`: represent (7) with the physical claim guarded.
3. `rotating_metric_limit`: prove vanishing at $\omega = 0$ with $\omega R/c$ control.
4. `helicity_invariant`: encode (12) as an ideal-deformation invariant.
5. `elsasser_relaxation`: prove relaxation of (14).
6. `lab_astro_functor`: preserve invariants (19) before lab results support astrophysical claims.
7. `orbital_movement_bridge`: connect drive and torque to lock thresholds.

10 Orders of magnitude: where the mechanism lives

It is worth fixing the numerical regimes the chain spans, because the functor of Section 7 is only as credible as the dimensionless overlap it asserts.

The Elsasser balance. For the Earth’s outer core ($B \sim 10^{-3}$ – 10^{-2} T internally, $\rho \approx 10^4$ kg m $^{-3}$, $\Omega \approx 7.3 \times 10^{-5}$ s $^{-1}$), the conventional Elsasser number sits at order unity—the canonical empirical motivation for ID0040’s universality claim. Laboratory dynamo experiments (liquid-sodium devices of the Riga/Karlsruhe/Madison class) reach the same order-unity balance at metre scale: eleven orders of magnitude below the planetary radius, yet on the same attractor. This is precisely the collapse predicted by (14) and the invariant-preservation rule (19): the attractor is indexed by Λ , not by size.

The moment scaling. The registry law (17) is the framework’s sharpened version of the long-observed astrophysical correlation between magnetic moment and angular momentum (the “magnetic Bode law” family), which spans planets, stars, and—with larger scatter—compact objects, across roughly twenty orders

of magnitude in $MR^2\Omega$. The framework’s falsifiable surplus is the claim that the residual scatter is structure (regime changes of the attractor), not noise; the sweep protocol inherited from Paper II (ID0144) applies verbatim with $\mu/(MR^2\Omega)$ as the order parameter.

The relativistic corrections. At laboratory rim speeds, $\omega R/c \sim 10^{-4}$ (ID10145), the rotating-metric correction (9) enters at 10^{-8} of the flat metric—small but within reach of interferometric phase measurements of the ID10264 suite; satellite-verified frame dragging (Lense–Thirring, confirmed at the Earth scale) anchors the $\omega R \ll c$ end of the same curve, while the $\omega R \rightarrow c$ gravitomagnetic onset (ID10272) remains the conjectural far end. The chain thus has one foot on confirmed physics, one foot on a guarded extrapolation, and a continuous dimensionless dial— $\omega R/c$ —connecting them.

11 Demarcation and predictions

Established physics: rigid-body and symplectic mechanics on $T^*SO(3)$; magnetohydrodynamic helicity and its ideal conservation; the Elsasser number as a dynamo balance diagnostic; frame dragging and gravitomagnetism in the weak-field limit; dimensional analysis and dynamical similarity; tidal capture into spin–orbit resonance.

Framework hypotheses, deliberately guarded: the gyromagnetic genesis claim (ID0085) that astrophysical angular momentum is *predominantly* generated rather than inherited; the compressed-dipole correction in (7); the universal constancy of k in (17); and the identification of the acoustic drive (20) with the enclosure modes of Papers I–II.

Falsifiable consequences. (i) *Moment–rotation scaling:* bodies spanning laboratory to stellar scales should collapse onto the single axis $\mu/(MR^2\Omega) = k$; strong systematic deviation breaks ID0044. (ii) *Elsasser convergence:* driven rotating magnetized systems should relax toward $\Lambda = 1$ with the first-order response (20); measured relaxation that is not exponential in $\Lambda - 1$ breaks ID0038. (iii) *Accumulated field:* the observable suite of ID10264 predicts a secular drift $g_{\text{acc}}/g_{\text{Newton}} = 1 + \alpha t + \beta t^2$; bounding α, β bounds the accumulation principle directly, anticipating the tests of Paper V. (iv) *Helicity transfer:* the predicted channel $\mathcal{H}_{\text{mag}} \rightarrow \mathcal{H}_{\text{KK}}$ couples the helicity invariant to the Kaluza–Klein tower, testable in principle through the resonance shift $\delta\omega/\omega = m_{\text{KK}}c/(R_{\text{KK}}\omega_0)$.

12 Conclusion: the accumulation principle, generalized

The movement mechanism is in hand. A typed state space (4) with coupled rotation–gauge symmetry; a guarded effective moment; a rotating metric correction vanishing with ω ; a topological invariant (12) that does the bookkeeping; an attractor (14) at which magnetic and rotational energy equipartition; a functor that disciplines every lab-to-cosmos inference; an acoustic gateway (20) through which the enclosure drives the system; and terminal capture into the rational locks of Paper II.

At its center sits one inversion: rotation as *accumulated torque history*, Eq. (3). Once movement is the integral of its own history, a question forces itself: is gravity any different? The registry’s answer—ID7214’s accumulated-force correction to Newtonian acceleration, ID10264’s accumulated-field observable already met above—is that gravitation is the same principle wearing geometric clothing: not a fundamental field, but the integrated response of mass-energy transport, with the Newtonian limit recovered as the far-field of an accumulation. Establishing that claim, with its Einstein-level closure $\nabla^2\Phi = 4\pi G_{\text{eff}}\rho$, is the task of Paper V.

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